ASSIGNMENT -01

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1. What is Full Stack Development? Explain the term Full Stack Development and the role of a Full Stack Developer.

Full Stack Development refers to the development of both the **client-side (front-end)** and **server-side (back-end)** of a web application. It involves working with user interfaces, server logic, databases, and APIs to create a complete application.

A **Full Stack Developer** is a software developer who has knowledge of front-end technologies (HTML, CSS, JavaScript), back-end technologies (Java, Python, Node.js, etc.), and databases (MySQL, MongoDB, etc.). They can design, develop, test, and maintain an entire web application from start to finish.

2. What is Frontend? Explain Frontend and list some technologies used to build it

Frontend is the part of a website or web application that users see and interact with directly. It is responsible for the layout, design, buttons, forms, navigation, and overall user experience.

Technologies used in Frontend Development:

- **HTML** – Creates the structure of web pages.
- **CSS** – Styles and designs the web pages.
- **JavaScript** – Adds interactivity and dynamic behavior.
- **React** – JavaScript library for building user interfaces.
- **Angular** – Frontend framework for web applications.
- **Vue.js** – Progressive JavaScript framework for UI development.

3. What is Backend? Explain Backend and mention its main responsibilities

Backend is the server-side part of a web application that works behind the scenes. It handles the application's logic, processes user requests, manages databases, and ensures that data is sent correctly to the frontend.

Main Responsibilities of Backend:

- Processing user requests.
- Managing and storing data in databases.
- Implementing business logic.
- Handling user authentication and authorization.
- Creating and managing APIs.
- Ensuring security and performance of the application.

4. What is a Database? Why do websites need databases? Give examples of data stored in a database.

Database is an organized collection of data that can be stored, managed, and retrieved efficiently by a computer system.

Why do websites need databases?
Websites use databases to store and manage information permanently. They help in retrieving, updating, and organizing data quickly whenever users interact with the website.

Examples of data stored in a database:

- User account details (name, email, password)
- Product information
- Customer orders
- Payment records
- Comments and reviews
- Booking and registration details

5. What is DNS? Explain how DNS helps users access websites.

DNS (Domain Name System) is a system that translates human-readable domain names (like **google.com**) into IP addresses that computers use to identify websites on the internet.

How DNS helps users access websites:
When a user enters a website name in a browser, DNS finds the corresponding IP address of that website. The browser then uses that IP address to connect to

the web server and load the website. This allows users to access websites using easy-to-remember names instead of numerical IP addresses.

6. What is an IP Address? Why are IP addresses important in networking?

IP Address (Internet Protocol Address) is a unique numerical identifier assigned to each device connected to a network or the internet. It helps identify and locate devices for communication.

Why are IP addresses important in networking?

- They uniquely identify devices on a network.
- They enable communication between devices.
- They help route data to the correct destination.
- They are essential for accessing websites and internet services.

Example: 192.168.1.1 (IPv4) or 2001:db8::1 (IPv6).

7. What is Client-Server Architecture? Explain how a client and server communicate with each other.

Client-Server Architecture is a network model in which a **client** (such as a web browser or mobile app) sends requests for services or data, and a **server** processes those requests and sends back the required response.

How a client and server communicate:

1. The client sends a request to the server over a network.
2. The server receives and processes the request.
3. The server accesses data or performs the required task.
4. The server sends a response back to the client.
5. The client displays the received information to the user.

Example: When you open a website, your browser (client) requests the webpage from a web server, and the server sends the webpage data back to the browser.

8. What is an API? Explain the role of APIs in web applications.

API (Application Programming Interface) is a set of rules that allows different software applications to communicate and exchange data with each other.

Role of APIs in Web Applications:

- Enable communication between the frontend and backend.
- Allow applications to access data and services from other systems.
- Help integrate third-party services such as payment gateways, maps, and social media.
- Facilitate secure and efficient data exchange.

Example: A weather app uses an API to fetch weather data from a weather service and display it to users.

9. What are HTTP Methods? Explain the purpose of: • GET • POST • PUT • DELETE

HTTP Methods are actions used by clients and servers to communicate and perform operations on data in a web application.

- **GET** – Retrieves data from the server.
- **POST** – Sends new data to the server to create a resource.
- **PUT** – Updates existing data on the server.
- **DELETE** – Removes data from the server.

Example:

- GET → Fetch user details.
- POST → Create a new user account.
- PUT → Update user information.
- DELETE → Delete a user account.

Developer Tools (DevTools) are built-in browser tools that help developers inspect, debug, test, and optimize web pages.

Three tabs in Developer Tools:

1. **Elements Tab** – Used to inspect and edit the HTML and CSS of a webpage.
2. **Console Tab** – Displays errors, warnings, and allows running JavaScript code.

3. **Network Tab** – Monitors network requests, API calls, and resource loading times.

These tools help developers identify and fix issues in web applications efficiently.

10. What are Developer Tools? Name any three tabs available in Developer Tools and explain their purpose.

Developer Tools (DevTools) are browser-based tools used by developers to inspect, debug, and test web applications.

Three tabs available in Developer Tools:

1. **Elements** – Used to view and edit the HTML and CSS of a webpage.
2. **Console** – Used to display errors, warnings, and run JavaScript code.
3. **Network** – Used to monitor network requests and analyze page loading performance.

These tools help developers find and fix problems in websites and web applications.

12. Draw the Complete Website Working Flow. User ↓ Browser ↓ DNS ↓ Server ↓ Database ↓ Server ↓ Browser ↓ Website Displayed Explain each step in one sentence.

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13. Student Registration System A student fills a registration form on a website. Explain: • What happens in the Frontend? • What happens in the Backend? • What happens in the Database?

When a student fills a registration form on a website:

Frontend:

- Displays the registration form to the student.
- Accepts input such as name, email, and password.
- Validates the entered data and sends it to the backend.

Backend:

- Receives the form data from the frontend.
- Validates and processes the data.
- Checks for errors such as duplicate email IDs.
- Sends the data to the database for storage.

Database:

- Stores the student's information permanently.
- Organizes the data in tables for future access and updates.
- Returns confirmation to the backend after successful storage.

14. Amazon Search Scenario A user searches for "Laptop" on Amazon. Explain the complete flow from entering the search term to displaying the results. Use the following terms in your answer: • Browser • Server • Database • Response

When a user searches for **"Laptop"** on Amazon:

1. The user enters **"Laptop"** in the **Browser** and clicks Search.
2. The browser sends a search request to the **Server**.
3. The server processes the request and searches the **Database** for laptop-related products.
4. The database returns the matching product information to the server.
5. The server prepares a **Response** containing the search results.
6. The response is sent back to the browser.
7. The browser displays the list of laptops to the user.

15. Website Observation Activity Open any website and inspect it using Developer Tools (F12). Write: • Website Name • One observation from the Elements Tab • One observation from the Network Tab • One observation from the Console Tab Reflection Questions

Website Observation Activity

Website Name: Google

Observation from Elements Tab:

- The webpage structure is built using HTML elements such as <div>, <form>, and <input> tags.

Observation from Network Tab:

- Multiple requests are sent to the server to load images, scripts, and stylesheets when the page opens.

Observation from Console Tab:

- The Console displays JavaScript messages, warnings, and errors generated by the webpage.

Reflection Questions

1. Why is the Elements tab useful?

- It helps developers inspect and modify the HTML and CSS of a webpage.

2. Why is the Network tab important?

- It shows how resources are loaded and helps identify performance issues.

3. What can be checked in the Console tab?

- JavaScript errors, warnings, and debugging information.

4. How do Developer Tools help web developers?

- They help inspect, debug, test, and optimize websites efficiently.

16. In your own words, explain how a website works from the moment a user enters a URL until the webpage appears on the screen.

When a user enters a **URL** in the browser and presses Enter, the browser first uses **DNS** to find the website's IP address. The browser then sends a request to the **server** hosting the website. The server processes the request, retrieves the required data from the **database** if needed, and sends a **response** back to the browser. The browser receives the response, interprets the HTML, CSS, and JavaScript files, and displays the webpage on the screen for the user to view and interact with.

17. Explain the difference between Frontend, Backend, and Database using a real-world example of your choice (Amazon, Instagram, Zomato, Netflix, etc.).

Difference Between Frontend, Backend, and Database (Using Amazon as an Example)

Frontend:
The frontend is the part of Amazon that users see and interact with, such as the homepage, search bar, product listings, and shopping cart.

Backend:
The backend processes user requests, handles business logic, manages orders, and communicates with the database. For example, when a user searches for a laptop, the backend processes the search request.

Database:
The database stores information such as product details, user accounts, order history, and payment records.

Example Flow:
When a user searches for a laptop on Amazon:

1. The **Frontend** collects the search term entered by the user.
2. The **Backend** receives the request and searches for matching products.
3. The **Database** provides the product information.
4. The **Backend** sends the results back to the **Frontend**.
5. The **Frontend** displays the list of laptops to the user.